Suryaveer Singh

716-295-3000 | suryaveer01@gmail.com | linkedin.com/in/suryaveer01 | github.com/suryaveer01

Professional Experience

Senior Robotics Engineer Dec 2024 – Present

Divine IT Services

- Architected a full-stack autonomy framework for industrial electric forklifts using **ROS2**, implementing a system of **Composable Nodes** to optimize CPU utilization and reduce inter-process communication latency
- Engineered a high-precision navigation stack leveraging **Nav2**, integrating **Extended Kalman Filters (EKF)** to fuse visual and wheel odometry for robust localization in dynamic warehouse environments
- Developed an advanced perception-actuation pipeline using **ZED Depth Cameras** and **AprilTag** visual fiducials; to facilitate sub-centimeter accuracy during autonomous pallet engagement
- Designed a hybrid Computer Vision workflow utilizing **Synthetic Data Generation (SDG)** and real-world datasets to train robust **Neural Networks** for multi-class bin detection across varying illumination profiles
- Optimized robotic manipulation sequences by synchronizing high-level motion planners with low-level **PLC motor controllers**, ensuring deterministic and smooth velocity ramps for heavy-load operations

Robotics Engineer May 2024 – Dec 2024

Setton Farms

- Contributed to the development of articulated **UGV**, utilizing **ROS** for sensor fusion, and navigation
- Developed a **robotic manipulator** with object detection to perform efficient **pick-and-place** operations

Software Engineer Jan 2016 – June 2022

Unisys

- Engineered 2 migration utilities for 300 million records, implementing event-driven architecture with **WebMQ**
- Developed an on-premises passenger facilitation solution at airports with **Identity verification** using iris/face
- Created a secure Remote Working Kit on **Azure Cloud Services**, for robust authentication and streamlined software distribution, enhancing remote work safety

Technical Skills

Programming Languages: Python, C/C++, Java(Spring Boot), SQL (Postgres), MATLAB
Frameworks/Libraries: ROS, ROS2, Issac-sim, Gazebo, RVIZ, moveIt, PyTorch, TensorFlow, TensorRT, Triton, ray, Stable Baselines3 (SB3)
Algorithms: Motion Planning, Path Planning, SLAM, Localization, 3-D Mapping, Obstacle Avoidance, Controls, DL/RL (Learning) State Estimation/Prediction, Optimization, Kinematics/Dynamics, Vision, tfs
Developer Tools: Git, Bitbucket, Tensorboard, Jenkins, Docker, Kubernetes, Azure

Education

University at Buffalo, The State University of New York

Masters of Science in Engineering Science Focus in AI and Robotics

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering

Projects

Motion Planning | *Python, Pytorch, Deep Learning, ROS, OpenCV, rosbag*

- Robotic Path Planning: Rapidly-exploring random trees(RRT) implementation on image maps using OpenCV
- SLAM: Successfully implemented an autonomous Evader-Mapper on an F1TENTH simulation simultaneously avoiding the obstacles and mapping the course using LIDAR data using python and rospy
- Robotic Controls: Implemented basic controls(PID and Pure Pursuit) for path following on an F1TENTH simulation with >90% accuracy in path
- Implemented point cloud clustering and downsampling for faster clustering using plyfile and open3d on python
- Visual Odometry: Calibrating the camera, extracting depth data, and implementing Stereo Visual Odometry using the vehicle's camera feed enabled accurate tracking and reliable analysis

Machine Learning | *Python, Stable Baselines3, Pytorch, Deep Learning, Reinforcement Learning*

- **Reinforcement Learning**: Successfully trained agents using **Stable Baselines3(SB3)** implementations of popular reinforcement learning algorithms such as **Deep Q-Network(DQN)**, **Proximal Policy Optimization(PPO)**, **Deep Deterministic Policy Gradient(DDPG)**, and **Soft Actor-Critic(SAC)** to achieve high-performance results within various OpenAI Gym environments
- **Mujoco**: Utilized **PPO** to train an agent for consistent walking in the Walker2D Mujoco environment
- **Deep Learning**: Attained high-accuracy classification and object detection in robotic applications using CNNs, and additionally implemented Ultralytics YOLOv8 on a custom dataset to compare performance